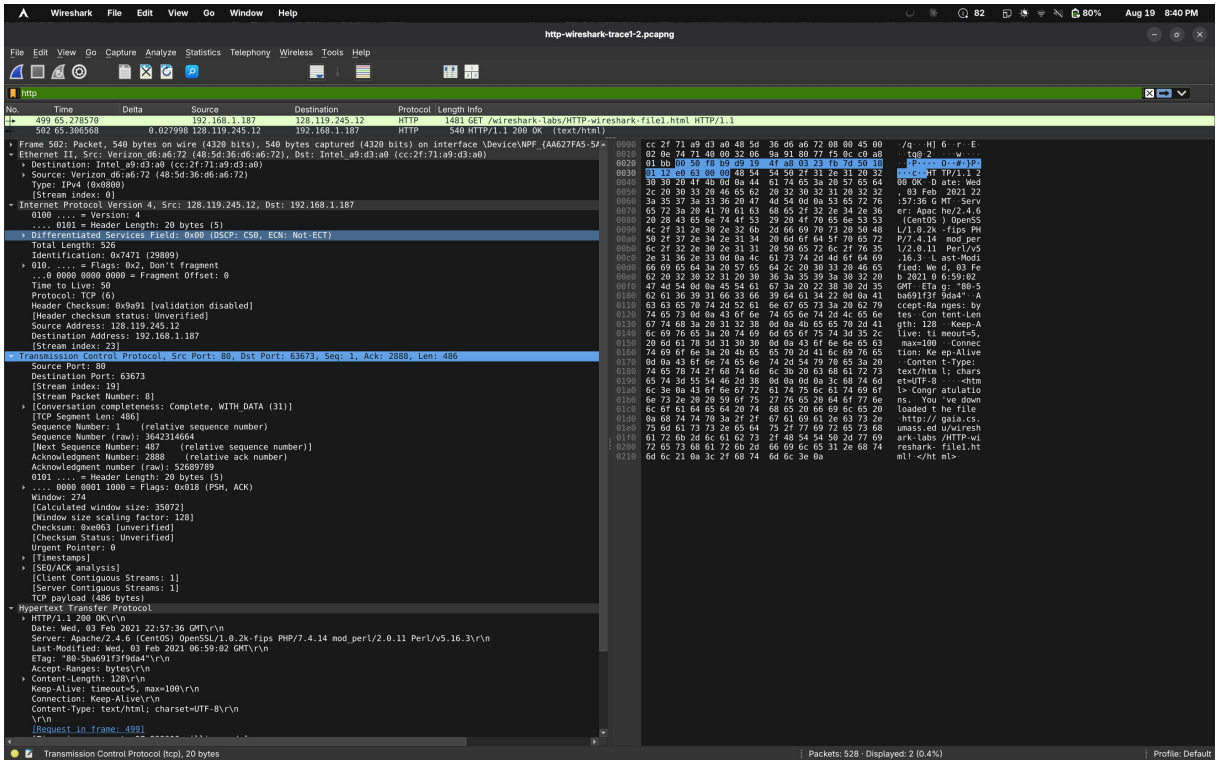
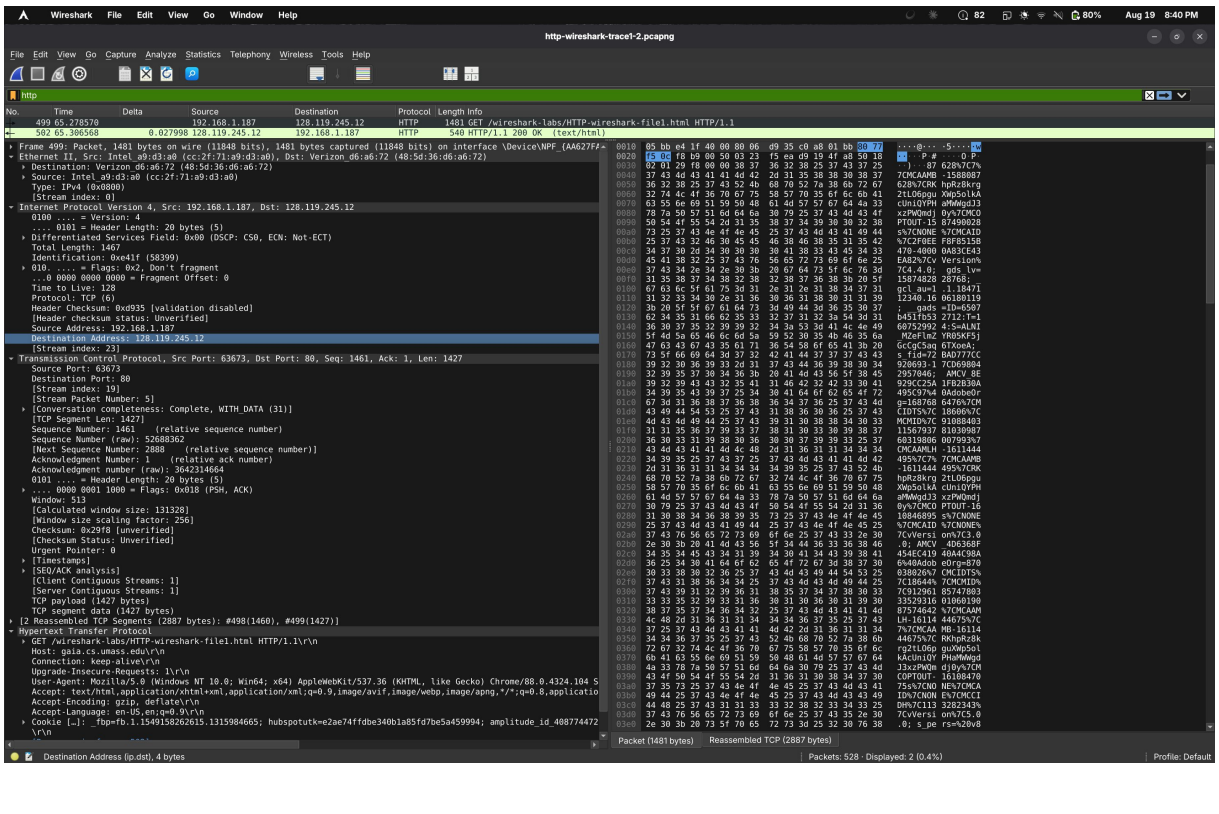
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Experiment 4 – Inspecting HTTP packets using Wireshark Packet Sniffer

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I. The Basic HTTP Get/Response Interaction





1. Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?
Both the browser and server are running HTTP/1.1 — the GET request reads GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1, and the response reads HTTP/1.1 200 OK.

2. What languages (if any) does your browser indicate that it can accept to the server?
The request shows Accept-Language: en-US,en;q=0.9 — so English (US) is preferred, with generic English as a fallback.

3. What is the IP address of your computer? Of the gaia.cs.umass.edu server?
End System (source): 10.116.29.233
Server (destination): 128.119.245.12

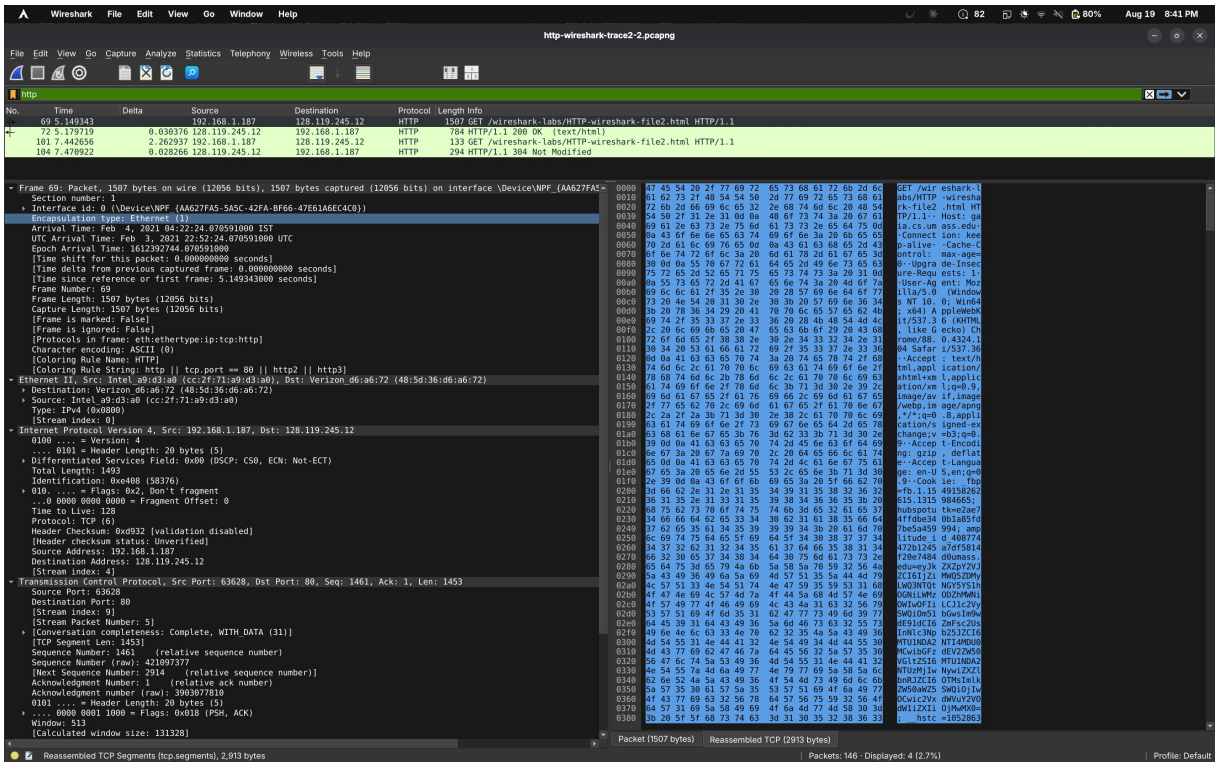
4. What is the status code returned from the server to your browser?
200 OK (visible in the packet list line for frame 325: HTTP/1.1 200 OK (text/html))

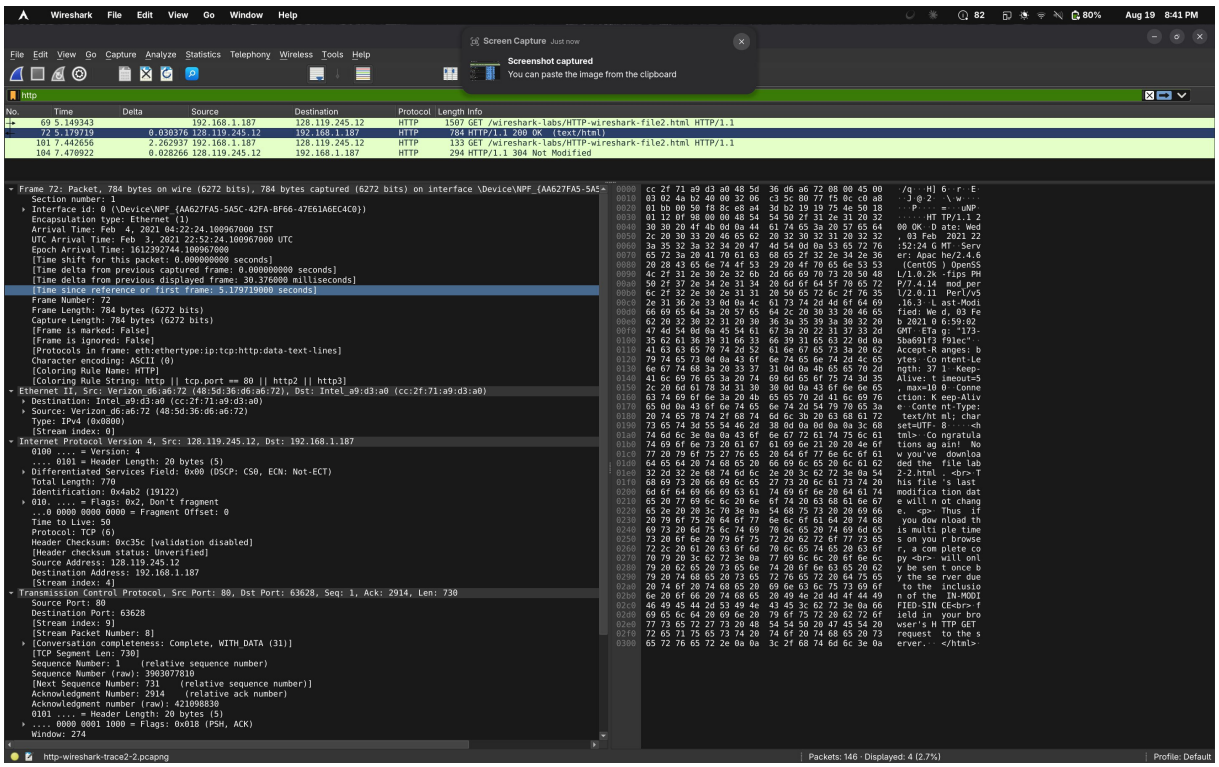
5. When was the HTML file that you are retrieving last modified at the server?
The file was last modified: Tue, 28 Oct 2025 05:59:01 GMT

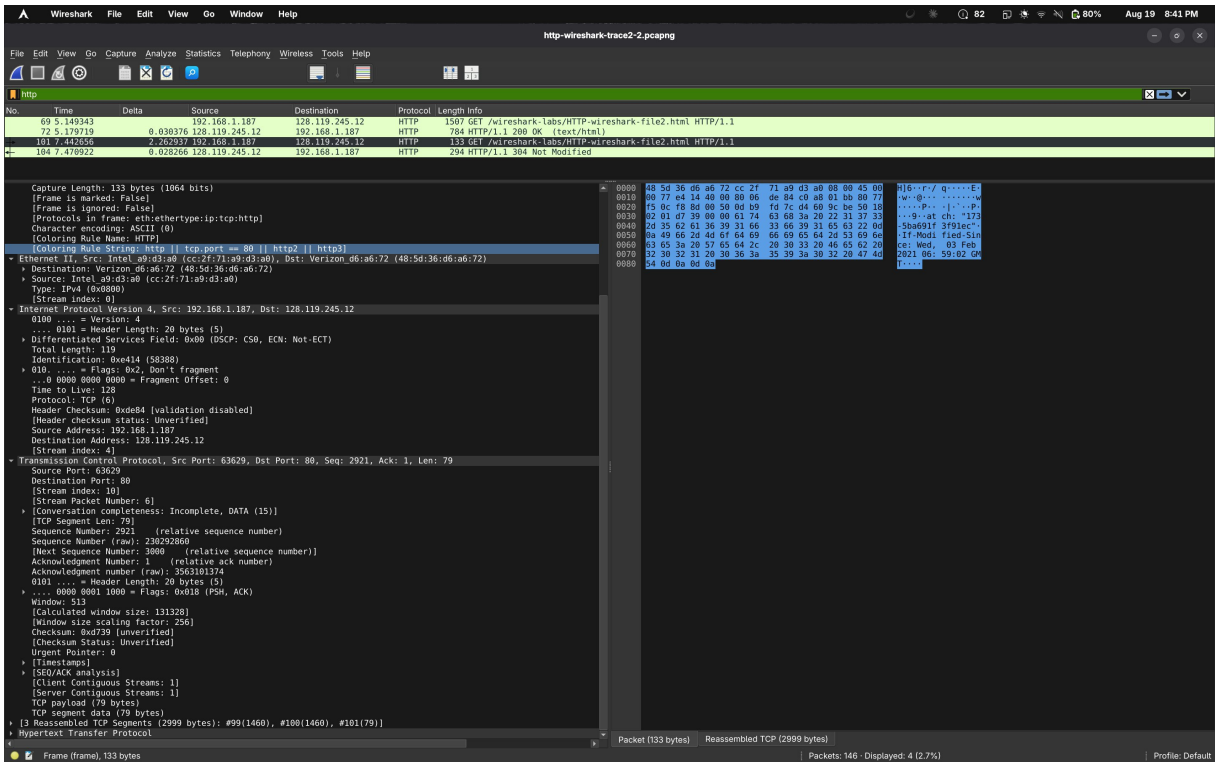
6. How many bytes of content are being returned to your browser?
128 bytes (Content-Length: 128), matching the actual HTML content shown at the bottom of the packet.

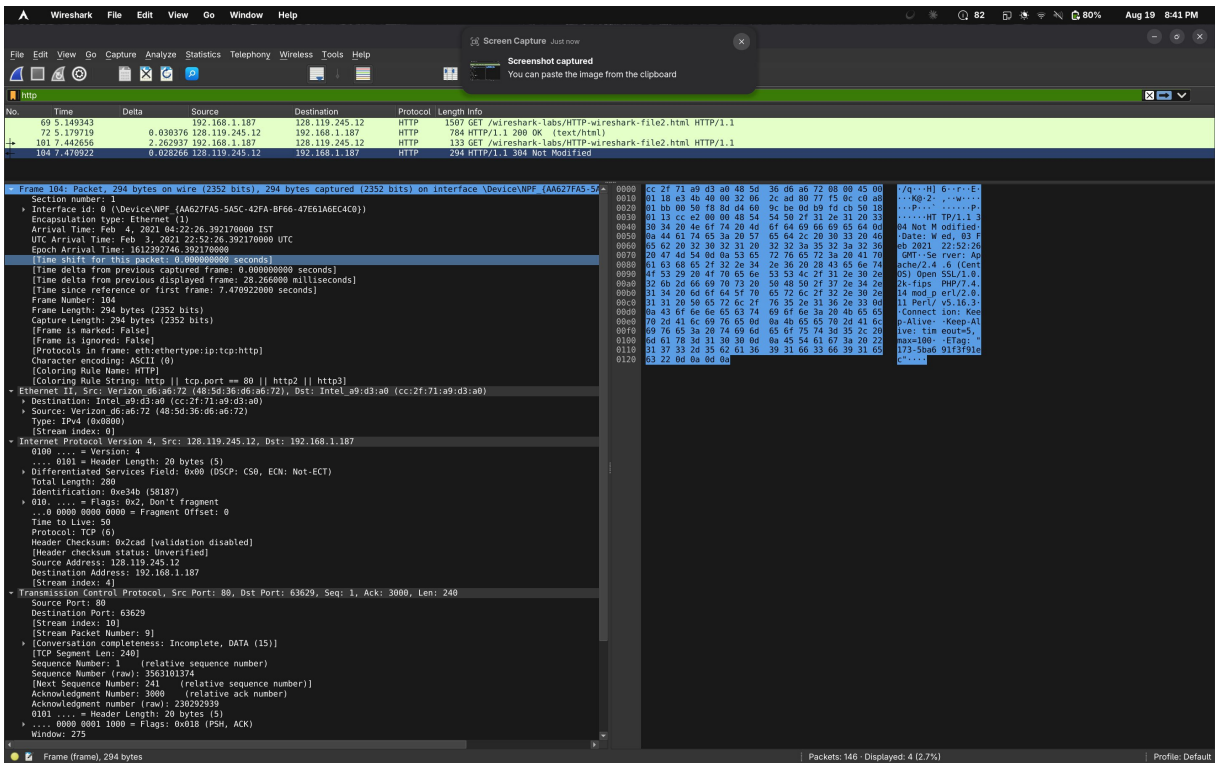
7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.
Yes, headers like Sec-GPC: 1 or Upgrade-Insecure-Requests: 1 are visible directly in the raw hex/ASCII dump but not in Wireshark's parsed output.

II. The HTTP CONDITIONAL GET/response interaction




Wireshark File Edit View Go Window Help
Aug 19 8:41 PM
Screen Capture Just now
Screenshot captured
You can paste the image from the clipboard
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Delta Source Destination Protocol Length Info
69 5.149343 192.168.1.187 128.119.245.12 HTTP 1507 GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
72 5.179719 0.030376 128.119.245.12 192.168.1.187 HTTP 784 HTTP/1.1 200 OK (text/html)
101 7.442656 2.262937 192.168.1.187 128.119.245.12 HTTP 133 GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
104 7.470922 0.028266 128.119.245.12 192.168.1.187 HTTP 294 HTTP/1.1 304 Not Modified
Frame 72: Packet, 784 bytes on wire (6272 bits), 784 bytes captured (6272 bits) on interface \Device\NPF_{AA627FA5-5A5
Section number: 1
Interface id: 0 (\Device\NPF_{AA627FA5-5A5C-42FA-BF66-47E61A6EC4C0})
Encapsulation type: Ethernet (1)
Arrival Time: Feb 4, 2021 04:22:24.100967000 IST
UTC Arrival Time: Feb 3, 2021 22:52:24.100967000 UTC
Epoch Arrival Time: 1612392744.100967000
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 0.000000000 seconds]
[Time delta from previous displayed frame: 30.376000 milliseconds]
[Time since reference or first frame: 5.179719000 seconds]
Frame Number: 72
Frame Length: 784 bytes (6272 bits)
Capture Length: 784 bytes (6272 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http || tcp.port == 80 || http2 || http3]
Ethernet II, Src: Verizon_d6:a6:72 (48:5d:36:d6:a6:72), Dst: Intel_a9:d3:a0 (cc:2f:71:a9:d3:a0)
Destination: Intel_a9:d3:a0 (cc:2f:71:a9:d3:a0)
Source: Verizon_d6:a6:72 (48:5d:36:d6:a6:72)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.1.187
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 770
Identification: 0x4ab2 (19122)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 50
Protocol: TCP (6)
Header Checksum: 0xc35c [validation disabled]
[Header checksum status: Unverified]
Source Address: 128.119.245.12
Destination Address: 192.168.1.187
[Stream index: 4]
Transmission Control Protocol, Src Port: 80, Dst Port: 63628, Seq: 1, Ack: 2914, Len: 730
Source Port: 80
Destination Port: 63628
[Stream index: 9]
[Stream Packet Number: 8]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 730]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 3903077810
[Next Sequence Number: 731 (relative sequence number)]
Acknowledgment Number: 2914 (relative ack number)
Acknowledgment number (raw): 421098830
0101 .... = Header Length: 20 bytes (5)
.... 0000 0001 1000 = Flags: 0x018 (PSH, ACK)
Window: 274
http-wireshark-trace2-2.pcapng
Packets: 146 · Displayed: 4 (2.7%)
Profile: Default


Wireshark File Edit View Go Window Help
Aug 19 8:41 PM
http-wireshark-trace2-2.pcapng
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http
No. Time Delta Source Destination Protocol Length Info
69 5.149343 192.168.1.187 128.119.245.12 HTTP 1507 GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
72 5.179719 0.030376 128.119.245.12 192.168.1.187 HTTP 784 HTTP/1.1 200 OK (text/html)
101 7.442656 2.262937 192.168.1.187 128.119.245.12 HTTP 133 GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
104 7.470922 0.028266 128.119.245.12 192.168.1.187 HTTP 294 HTTP/1.1 304 Not Modified
Capture Length: 133 bytes (1064 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:tcp:http]
Character encoding: ASCII (0)
[Coloring Rule Name: HTTP]
[Coloring Rule String: http || tcp.port == 80 || http2 || http3]
Ethernet II, Src: Intel_a9:d3:a0 (cc:2f:71:a9:d3:a0), Dst: Verizon_d6:a6:72 (48:5d:36:d6:a6:72)
Destination: Verizon_d6:a6:72 (48:5d:36:d6:a6:72)
Source: Intel_a9:d3:a0 (cc:2f:71:a9:d3:a0)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 192.168.1.187, Dst: 128.119.245.12
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 119
Identification: 0xe414 (58388)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0xde84 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.187
Destination Address: 128.119.245.12
[Stream index: 4]
Transmission Control Protocol, Src Port: 63629, Dst Port: 80, Seq: 2921, Ack: 1, Len: 79
Source Port: 63629
Destination Port: 80
[Stream index: 10]
[Stream Packet Number: 6]
[Conversation completeness: Incomplete, DATA (15)]
[TCP Segment Len: 79]
Sequence Number: 2921 (relative sequence number)
Sequence Number (raw): 230292860
[Next Sequence Number: 3000 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 3563101374
0101 .... = Header Length: 20 bytes (5)
.... 0000 0001 1000 = Flags: 0x018 (PSH, ACK)
Window: 513
[Calculated window size: 131328]
[Window size scaling factor: 256]
Checksum: 0xd739 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (79 bytes)
TCP segment data (79 bytes)
[3 Reassembled TCP Segments (2999 bytes): #99(1460), #100(1460), #101(79)]
Hypertext Transfer Protocol
Packet (133 bytes)
Reassembled TCP (2999 bytes)
Frame (frame), 133 bytes
Packets: 146 · Displayed: 4 (2.7%)
Profile: Default



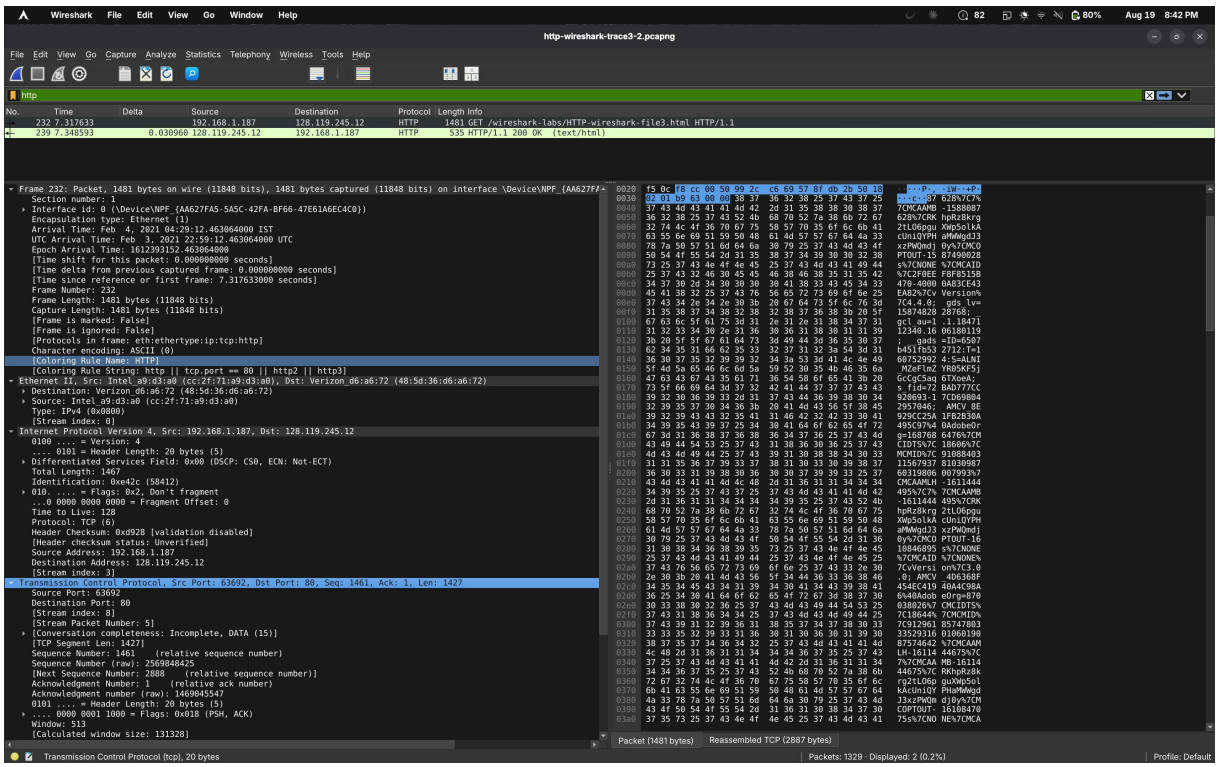
8. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?
Yes — frame 412 (the GET for /wireshark-labs/HTTP-wireshark-file2.html) already includes:
If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT
If-None-Match: "173-64231b67176b7"

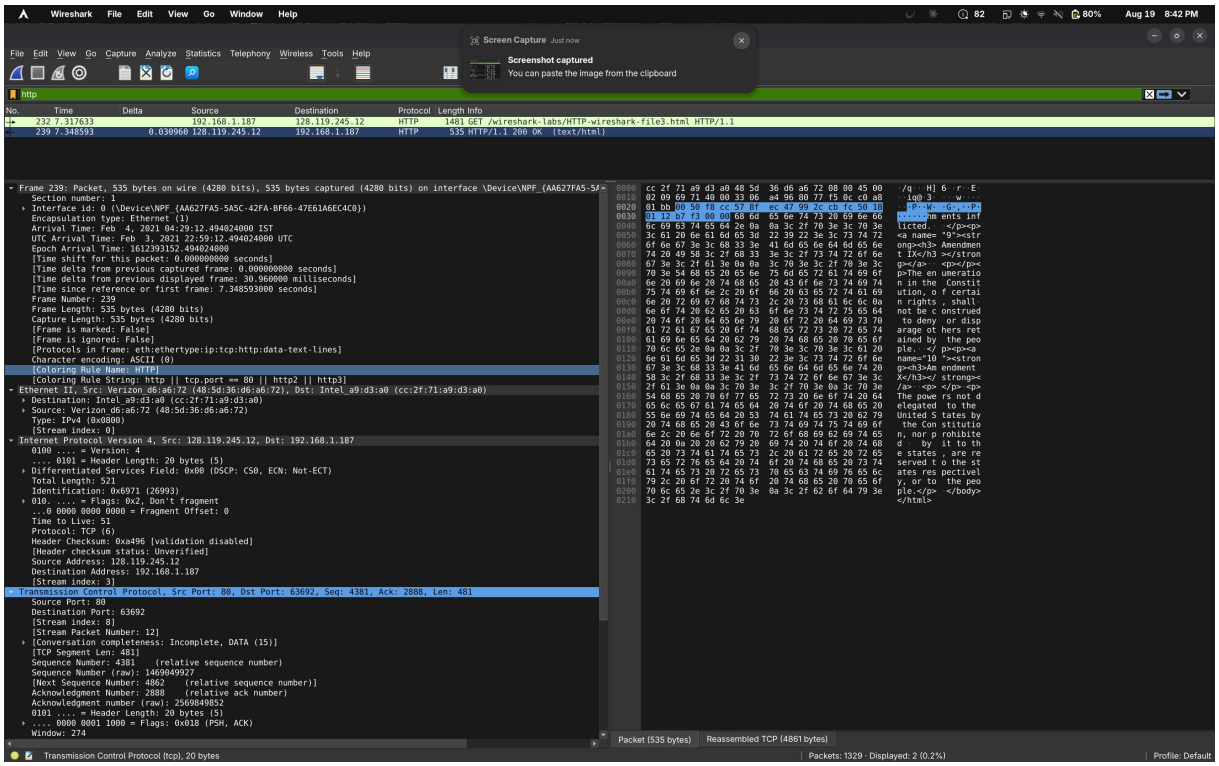
9. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?
No. The response (frame 414) is: HTTP/1.1 304 Not Modified. There is no body or File Data section in the response — only headers are visible. This means the cached copy is valid and the server did not resend the file.

10. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?
Yes — frame 412 contains:
If-Modified-Since: Tue, 28 Oct 2025 05:59:01 GMT

11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.
The response (frame 414) was: HTTP/1.1 304 Not Modified. The status code 304 means the cached copy is still current. There is no Content-Length header and no body in the response — only headers are visible. This confirms the browser used its cached copy of file2.html instead of the server re-sending the HTML.

III. Retrieving Long Documents





12. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill of Rights?

The browser sent one GET request for the Bill of Rights. It is in frame 19: GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1

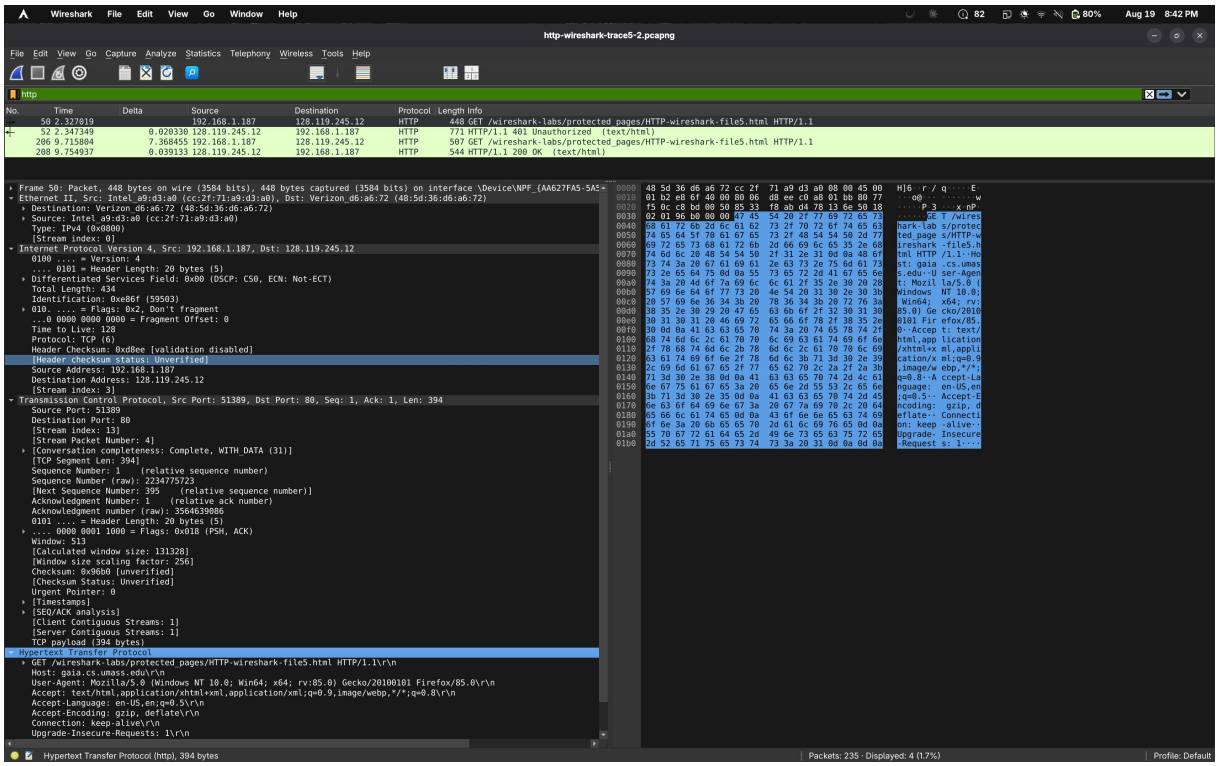
13. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

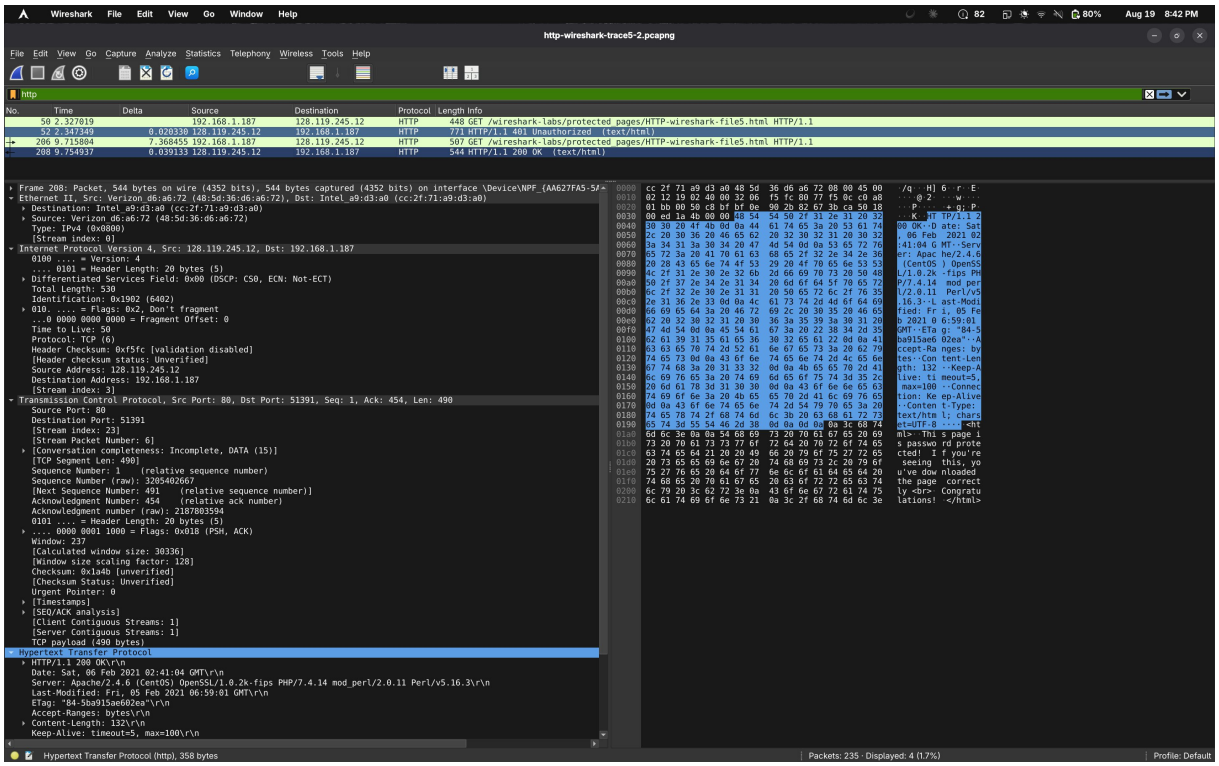
Frame 25 contains the response headers, HTTP/1.1 200 OK, and the reassembled body.

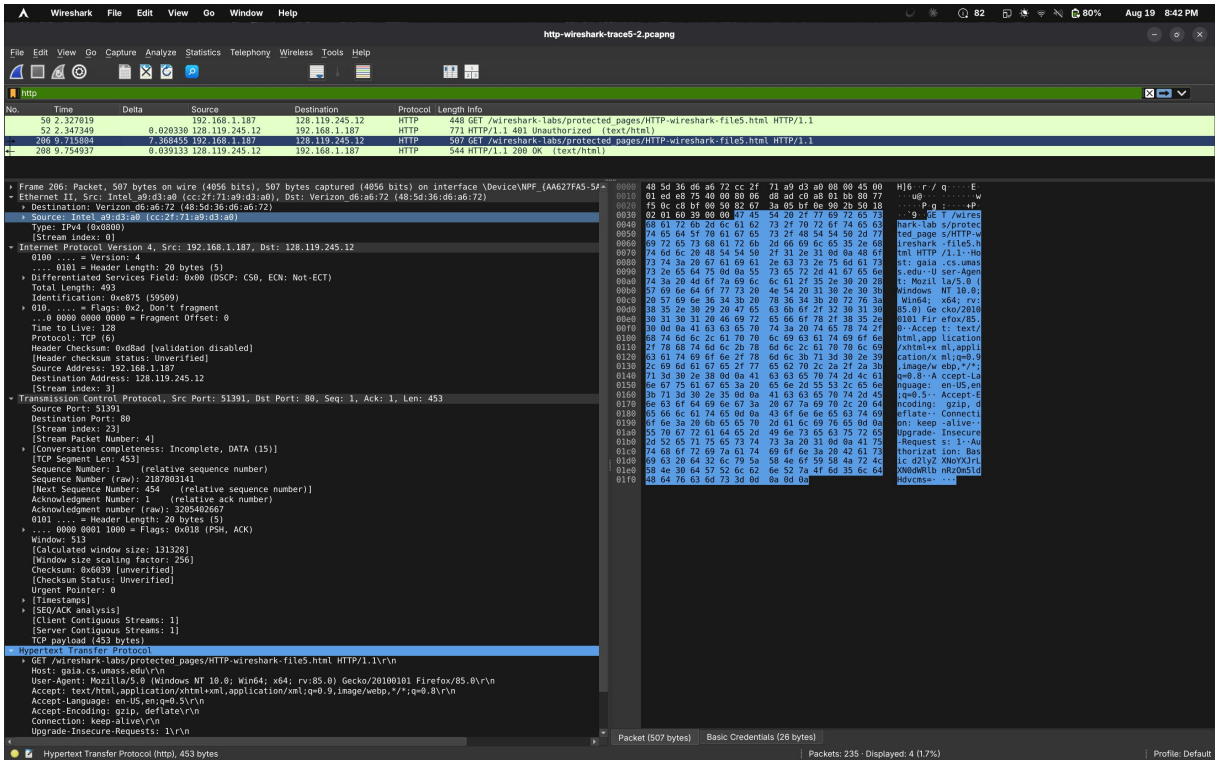
14. What is the status code and phrase in the response?
200 OK

15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?
4 segments: [4 Reassembled TCP Segments (4864 bytes): #22(1238), #23(1238), #24(1238), #25(1150)]

IV. HTML Documents with Embedded Objects







16. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

Your browser sent 4 GET requests total:

487 GET /wireshark-labs/HTTP-wireshark-file4.html → 128.119.245.12 (gaia.cs.umass.edu)

496 GET /pearson.png → 128.119.245.12 (gaia.cs.umass.edu)

500 GET /8E_cover_small.jpg → 2.56.99.24 (kurose.cslash.net)

506 GET /favicon.ico → 128.119.245.12 (gaia.cs.umass.edu)

Two embedded objects (pearson.png and 8E_cover_small.jpg) were referenced in the HTML and sent to two different servers.

17. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

In parallel. Evidence from the timestamps:

Frame 496: GET /pearson.png sent at 14.853523635s

Frame 500: GET /8E_cover_small.jpg sent at 15.006384037s

Frame 504: response (301 redirect) for pearson.png arrives at 15.087315185s

The second GET was sent before the response to the first arrived, proving parallel (pipelined/concurrent) downloading.